Forming Equations from Shapes

Exam code: 0607

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Forming equations from shapes

How do I form equations from shapes?

- You need to use all the information given on the diagram and any specific **properties** of that **shape**
- Common **2D shapes** that you should know properties for are
  - Triangles: **equilateral**, **isosceles**, **scalene**, **right-angled**
  - Quadrilaterals: **square**, **rectangle**, **kite**, **rhombus**, **parallelogram**, **trapezium**
- You may be asked about **perimeter**, **area** or **angles**
- You may be asked about **polygons**
  - **Regular** vs **irregular** polygons
  - **Interior** vs **exterior** angles
    - The **sum** of **interior** angles is **180(*n*−2)** for an *n*-sided polygon
- You may be asked about **angles** in **parallel lines**
  - **Alternative**, **corresponding** and **co-interior**
- You may be asked about **3D shapes** involving **surface area** and **volume**
  - **Prisms** have constant cross sections
    - Volume is cross-section area multiplied by length

Is there anything else that can help?

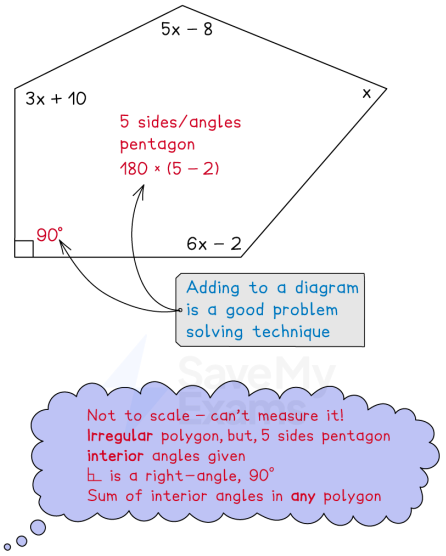
- **Sketch** a diagram if none is given
- **Split** up uncommon shapes into the **sum** or **difference** of common shapes
- Look out for important extra information
  - For example, a trapezium "with a line of symmetry"
- With **irregular** shapes, **assume** all angles and lengths are different (unless told otherwise)
- Put **brackets** around **algebraic expressions** when substituting them into geometric properties

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Sum of angles = 180(n − 2)
= 180(5 − 2)
= 180 × 3
= 540

3x + 10 + 5x − 8 + x + 6x − 2 + 90 = 540

15x + 90 = 540

15x = 450

x = 30°

A good alternative question here would be to find the smallest or largest angle (30°, 178° !)

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Examiner Tips and Tricks

Read the question carefully - does it want an angle? perimeter? total area? curved surface area? etc.

For surface area and volume questions, check the list of formulas given in the exam.

Worked Example

A rectangle has a length of $3x + 1$ cm and a width of $2x - 5$ cm.

Its perimeter is equal to 22 cm.

(a) Use the above information to find the value of x.

Answer:

The perimeter of a rectangle is 2 × length + 2 × width

$$2(3x + 1) + 2(2x - 5)$$

Expand the brackets

$$6x + 2 + 4x - 10$$

This perimeter is 22, so set this expression equal to 22

$$10x - 8 = 22$$

Solve this equation by adding 8 then dividing by 10

$$10x = 22 + 8$$
$$10x = 30$$
$$x = \frac{30}{10}$$
$$x = 3$$

$$x = 3$$

(b) Find the area of the rectangle.

Answer:

The area of a rectangle is its length multiplied by is width
Substitute the value of x from part (a) into the length and width given in the question

length is $3 \times 3 + 1 = 10$

width is $2 \times 3 - 5 = 1$

Find the area (multiply length by width)

10×1

Include the correct units for area

Area = 10 cm^2

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